

Ankush Jain

Backend Engineer

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SUMMARY

Backend engineer, five years on distributed systems for regulated healthcare and imaging - EHR integration, cloud infrastructure, access control and AI pipelines. Owns systems from design to the CI/CD that ships them.

EXPERIENCE

Apra Labs

Software Engineer

May 2023 – Present

Engineering services across healthcare, medical devices and imaging; clients not named.

Clinical Trial Management Platform (Nov 2024 – Present)

- Built the integration hub's base adapter modules, turning each new EHR integration from a bespoke build into a repeatable pattern
- Designed and shipped a full RBAC system across two applications: admin-only endpoint restriction, per-role response filtering and pre-authorization checks
- Re-architected inter-service connectivity on managed pub/sub with async subscriptions, dead-letter handling and retry enhancement
- Cut the backend deploy pipeline from 26m33s to 18m07s (-32%)

Regenerative Medicine Data Platform (May 2026 – Present)

- Modelled clinical protocols and surveys on FHIR resources with OMOP/LOINC terminology; implemented change logs via FHIR provenance
- Drove requirements with the client directly - converting whiteboard sessions and slide feedback into wireframed workflows and user stories

Document AI / Handwriting Recognition Pipeline (May 2025 – Nov 2025)

- Cut the deployable image from 19 GB to 5.28 GB (-72%) through packaging design
- Designed the codebase so the third-party model packages and deploys with the system and its licence can be swapped on expiry, instead of requiring a rebuild

Headless Video Recording Platform (Apr 2024 – Oct 2024)

- Wrote the Node backend from scratch and connected it to Redis, the C++ core and the frontend, giving the system one consistent source of truth

AR Surgical Navigation (Aug 2023 – Apr 2024)

- Found a signed distance field computation dominating the pipeline; compute-once-and-reuse removed over 80% of its per-frame cost

ByteLearn

Software Engineer

May 2022 – Apr 2023

- Built out monitoring and log management: Graylog for structured logging, Sentry APM across the Flask and Node.js services
- Core maintainer of the open-source mathsteps library; added complex expression support and improved parsing accuracy

Deloitte USI

Business Technical Analyst (Software Developer)

Sep 2021 – May 2022

- Contributed to migrating a monolithic insurance premium service to AWS serverless (Lambda, Step Functions)
- Maintained the integration layer of an AWS Step Functions state machine, implementing API calls and Lambda functions

SKILLS

Languages: Python, Java, Kotlin, C++, Go, TypeScript/Node.js, SQL

Backend & data: Spring Boot, Ktor, REST, Server-Sent Events, RabbitMQ, GCP Pub/Sub, PostgreSQL, Redis, Event-driven architecture

Cloud & DevOps: GCP (Cloud Run), AWS (Lambda, Step Functions), Terraform, Docker, CI/CD, Observability (Sentry, Graylog)

Healthcare & security: HL7 FHIR, OMOP/LOINC, EHR integration, RBAC, Authentication & authorization

ML & imaging: Computer vision, Signal processing (FFT/DFT), OCR/HTR, Hyperspectral imaging

EDUCATION

Dr. SPM IIIT Naya Raipur — B.Tech, Computer Science and Engineering

2017 – 2021